

Numerical Problems

1. A generating station has a maximum demand of 30 MW a load factor of 0.6, a plant capacity factor of 0.48 and a plant use factor of 0.82. Find
 - (i) The daily energy produced
 - (ii) The reserve capacity of the plant
 - (iii) The maximum energy that could be produced daily if the plant were running all the time
 - (iv) The maximum energy that could be produced daily if the plant when running according to operational schedule were fully loaded.
2. A 60 MW power station has an annual peak load of 50 MW. The power station supplies loads having maximum demands of 20MW, 17 MW, 10 MW and 9 MW. The annual load factor is 0.45. Find
 - (i) Average Load
 - (ii) Energy supplied per year
 - (iii) Diversity factor
 - (iv) Demand Factor
3. At the end of a power distribution system, a certain feeder supplies three distribution transformers, each one supplying a group of consumers whose connected loads are as under

Transformer	Load	Demand factor	Diversity Factor
Transformer No. 1	10 kW	0.65	1.5
Transformer No. 2	12 kW	0.6	3.5
Transformer No. 3	15 kW	0.7	1.5

If the diversity factor among the transformers is 1.3, find the maximum load on the feeder.

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